

Diet Patterns, Food Safety, and Chronic Disease Risk in Domestic Dogs

ICS604 – Applied Data Science | Final Project Report Author: Maritza Medina | Date: Spring 2026

1. Introduction and Motivation

The relationship between diet and long-term health is actively debated in companion animal medicine. Dog owners increasingly choose among conventional commercial kibble, raw meat-based diets, and plant-based (vegan) formulations, often with limited evidence. While anecdotal claims abound, rigorous comparative data on chronic disease outcomes across diet types is scarce.

This project investigates whether diet type is associated with the prevalence of chronic diseases in domestic dogs using a large survey dataset. The analysis is motivated by two concerns: chronic diseases (cancer, heart disease, GI disorders) are leading causes of canine morbidity; and the pet food industry faces ongoing regulatory scrutiny, including FDA recalls and investigations into grain-free diets and dilated cardiomyopathy (DCM). Understanding how diet relates to disease burden has direct relevance for veterinary guidance.

2. Dataset Description

The dataset is open-access survey data from Knight et al. (2022) [1], hosted on the Open Science Framework (<https://osf.io/nbepu>). It contains **2,536 dog records** collected via an online survey. Each owner reported on one dog fed the same diet for at least one year. Key features include demographics (breed size, age, sex/neuter status), lifestyle (activity, therapeutic diet, medications), diet type (Conventional, Raw, Vegan), and 22 binary health conditions.

Preprocessing involved: (1) renaming 35 survey columns; (2) converting disease indicators to binary 0/1; (3) parsing age into a numeric variable; (4) collapsing diet into a three-group categorical and a binary commercial vs. non-commercial variable; and (5) creating a composite `any_chronic_disease` outcome (cancer, heart disease, allergy, or mobility issues). Rows missing age or diet were excluded, leaving **2,534 complete cases**.

The primary challenge was the cross-sectional, self-reported nature of the data, introducing potential recall bias. Additionally, raw-fed dogs are on average 2–3 years younger than vegan-fed dogs, making age adjustment essential.

3. Research Questions and Objectives

This project addresses five research questions, progressing from descriptive to predictive analysis:

1. **RQ1 (Descriptive):** Is there a statistically significant unadjusted association between diet type and the prevalence of selected chronic diseases?
2. **RQ2 (Comparative):** Do dogs fed home-made/raw diets differ from dogs fed commercial diets in chronic condition prevalence?
3. **RQ3 (Inferential):** After controlling for demographic and lifestyle factors, does diet type predict the adjusted odds of a dog having cancer?
4. **RQ4 (Predictive):** What variables are the strongest predictors of chronic disease risk?
5. **RQ5 (Exploratory):** What do FDA pet food recall trends reveal about contamination risks in the commercial supply?

Note: RQ1 provides an unadjusted picture across multiple diseases, while RQ3 narrows the focus to cancer and applies a full logistic regression model with covariates.

4. Methods and Approaches

RQ1: Chi-square tests of independence were applied to 10 primary disease indicators against the three-group diet variable. Chi-square is appropriate for categorical predictors and outcomes. **RQ2:** Two-group proportion comparison collapsed diet into a binary variable (Commercial vs. Non-commercial) and computed prevalence rates

and chi-square statistics. **RQ3:** Logistic regression modeled the binary cancer outcome as a function of diet type, age, breed size, and sex/neuter status, producing interpretable odds ratios (ORs). Age was included as a continuous covariate due to confounding. **RQ4:** Decision Tree and Random Forest classifiers predicted the composite `any_chronic_disease` outcome ($70/30$ split). Models were evaluated using precision, recall, F1-score, and ROC-AUC, focusing on recall for the minority class (~9% prevalence). **RQ5:** Descriptive time-series analysis of FDA Center for Veterinary Medicine (CVM) pet food recall data (2012–2023). This is explicitly exploratory and non-causal.

5. Results and Analysis

RQ1: Chi-square tests revealed significant associations for four conditions. Commercial-fed dogs had notably higher rates of GI issues (9.43% vs. 4.22% raw vs. 4.46% vegan, $p < 0.001$), obesity (4.46% vs. 1.45% vs. 2.08%, $p < 0.001$), mobility problems (6.58% vs. 3.49% vs. 4.76%, $p < 0.01$), and dental issues (5.48% vs. 2.77% vs. 3.57%, $p < 0.01$). Cancer prevalence was not significantly different across groups.

RQ2: When collapsed to a binary variable, commercial-fed dogs had significantly higher rates of obesity (4.46% vs. 1.63%, $p < 0.001$), GI issues (9.43% vs. 4.29%, $p < 0.001$), and the composite any-chronic-disease outcome (10.53% vs. 7.63%, $p = 0.015$).

RQ3: The logistic regression model found that **age was the only statistically significant predictor of cancer** (OR = 1.20 per year, 95% CI: 1.11–1.30, $p < 0.001$). Neither raw meat-based diet ($p = 0.319$) nor vegan diet ($p = 0.987$) was significantly associated with cancer risk after adjustment. Small breed size was associated with significantly lower cancer odds ($p = 0.004$).

RQ4: The Random Forest classifier achieved 80% accuracy and a ROC-AUC of 0.654. Recall for the chronic disease class was limited (0.30) due to class imbalance. Feature importance rankings identified **age** as the dominant predictor, followed by **veterinary visit frequency** and **medication use**. Diet type features ranked lower.

RQ5: Between 2012 and 2023, the FDA CVM issued 370 pet food recalls. The peak year was 2019 (47 recalls), coinciding with the FDA's DCM investigation. Contamination-related recalls accounted for ~64% of all events. No causal connection to the survey data is inferred.

6. Discussion and Insights

The most consistent finding (RQ1, RQ2) is that commercial-fed dogs have significantly higher rates of GI issues, obesity, mobility problems, and dental disease compared to raw or vegan-fed dogs. However, raw-fed dogs are on average 2–3 years younger. The adjusted analysis (RQ3) confirms that once age is controlled, diet type does not significantly predict cancer risk, suggesting unadjusted differences may be attributable to age confounding.

The RQ4 finding that age, vet visits, and medication use dominate the predictive model is noteworthy. Vet visits and medication use are likely proxies for pre-existing illness, highlighting a limitation of cross-sectional data.

Limitations: (1) Health outcomes are owner-reported, not veterinary-confirmed; (2) cross-sectional design prevents causal inference; (3) the sample is self-selected; (4) cancer and heart disease are rare (~2%), limiting statistical power; and (5) FDA recall data and survey data are independent.

Future extensions could include longitudinal follow-up, linkage to veterinary records, breed-stratified analyses, and formal mediation analysis to decompose age confounding.

7. Use of AI Tools

This project used **Manus AI** for assistance with code scaffolding, notebook structure design, and report drafting. Manus generated the initial notebook template, suggested a robust column-rename helper function for Unicode issues, and drafted sections of this report. All analytical decisions, research question formulation, interpretation of results, and final edits were made by the student.

8. Conclusion

This project examined dog diet type and chronic disease prevalence. Unadjusted analyses revealed commercial-fed dogs have higher rates of GI issues, obesity, mobility problems, and dental disease. However, adjusted logistic regression found age — not diet — is the primary driver of cancer risk, and predictive modeling confirmed age and healthcare-access variables dominate over diet in predicting

chronic disease. The secondary FDA recall analysis documented a peak in 2019 driven by contamination, but no causal link to disease patterns was established. Results suggest diet associations are substantially confounded by age; longitudinal data is essential to establish causal dietary influences.

References

- [1] Knight A, Huang E, Rai N, Brown H (2022). *Vegan versus meat-based dog food: Guardian-reported indicators of health*. PLOS ONE 17(4): e0265662. <https://doi.org/10.1371/journal.pone.0265662>
- [2] Barrett-Jolley R, German AJ (2024). *Reanalysis of a survey of dog owners' perceptions of health outcomes associated with diet type*. PLOS ONE. <https://doi.org/10.1371/journal.pone.0280173>
- [3] U.S. Food & Drug Administration. *FDA Investigation into Potential Link between Certain Diets and Canine Dilated Cardiomyopathy*. <https://www.fda.gov/animal-veterinary/outbreaks-and-advisories/fda-investigation-potential-link-between-certain-diets-and-canine-dilated-cardiomyopathy>
- [4] U.S. FDA Center for Veterinary Medicine. *Recalls & Withdrawals*. <https://www.fda.gov/animal-veterinary/safety-health/recalls-withdrawals>
- [5] American Veterinary Medical Association (AVMA). *Raw Pet Foods*. <https://www.avma.org/resources-tools/avma-policies/raw-or-undercooked-animal-source-protein-companion-animal-diets>

Supplementary Materials: Figures and Tables

Table 1: RQ1 — Chi-Square Test Results: Diet Type vs. Disease Prevalence
(*** p<0.001, ** p<0.01, * p<0.05, ns = not significant; yellow = significant)

Condition	Commercial (%)	Raw/Home-prep (%)	Vegan (%)	Chi²	p-value	Sig.
Cancer	1.83	2.17	2.08	0.34	0.8457	ns
Heart	2.05	0.96	2.68	5.28	0.0713	ns
Allergy	0.95	0.84	0.00	3.16	0.2063	ns
Obesity	4.46	1.45	2.08	16.80	0.0002	***
GI Issues	9.43	4.22	4.46	25.37	<0.0001	***
Mobility	6.58	3.49	4.76	9.99	0.0068	**
Dental	5.48	2.77	3.57	9.70	0.0078	**
Skin	6.21	5.54	4.76	1.19	0.5511	ns
Kidney	0.51	0.84	0.89	1.14	0.5657	ns
Hormonal	1.54	0.72	1.19	2.82	0.2437	ns

Figure 1: Disease Prevalence (%) by Diet Type — Three-Way Comparison

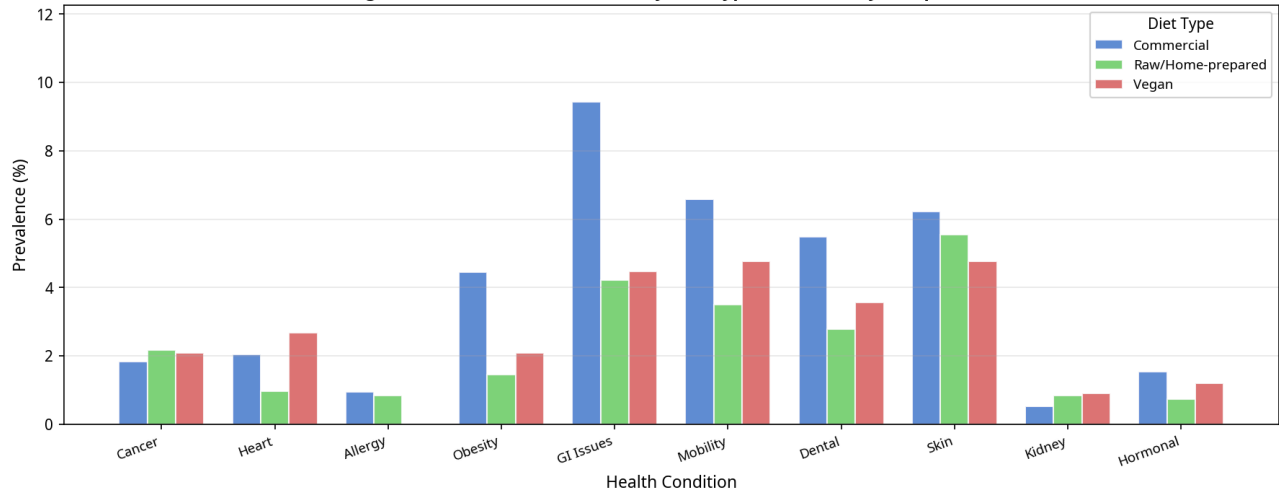


Figure 3: RQ3 — Adjusted Odds Ratios for Cancer Risk (Reference: Commercial Diet)

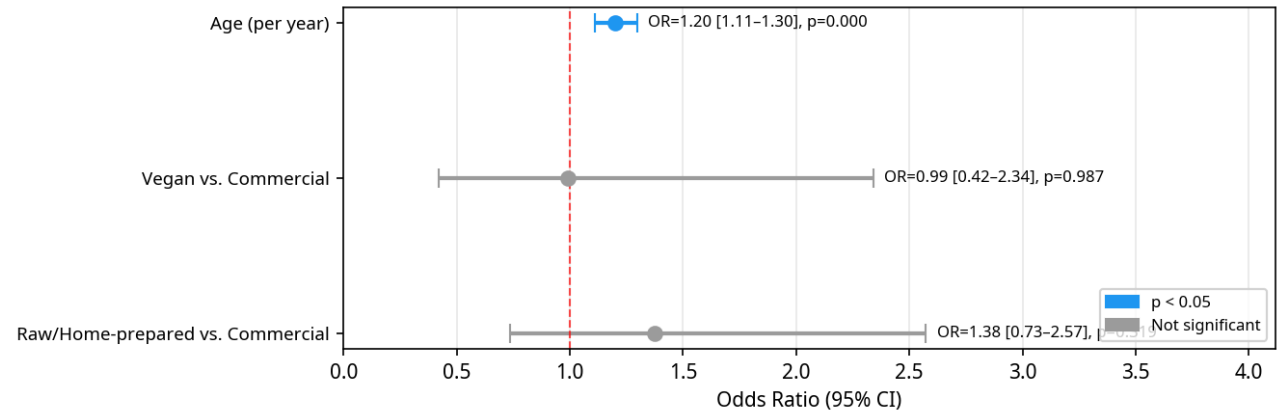


Table 2: RQ3 — Logistic Regression: Adjusted Odds Ratios for Cancer Risk
(Reference: Commercial diet; yellow = statistically significant)

Predictor	OR	95% CI	p-value	Sig.
Raw/Home-prep vs. Commercial	1.38	[0.73 – 2.57]	0.319	ns
Vegan vs. Commercial	0.99	[0.42 – 2.34]	0.987	ns
breed_size_Large (22.7–40.5 kg or 50–88 lb)	0.40	[0.13 – 1.23]	0.109	ns
Medium Breed vs. Large	0.37	[0.12 – 1.13]	0.080	ns
Small Breed vs. Large	0.12	[0.03 – 0.50]	0.004	**
breed_size_Toy (≤ 4.1 kg or ≤ 9 lb)	0.00	[0.00 – inf]	0.999	ns
Female vs. Male	1.07	[0.36 – 3.20]	0.904	ns
sex_neuter_Male, castrated	0.53	[0.17 – 1.66]	0.274	ns